ITA0615 - MACHINE LEARNING

DECISION TREE LEARNING

A simple way to make decisions using data

Technical Presentation | CO1

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1. What Is a Decision Tree?

A tree-shaped model used for classification and decision-making

- It learns simple rules from examples in a dataset.
- Each internal node asks a question about the data.
- Branches show possible answers; leaf nodes show the final result.

Tree structure

ROOT NODE
The first question

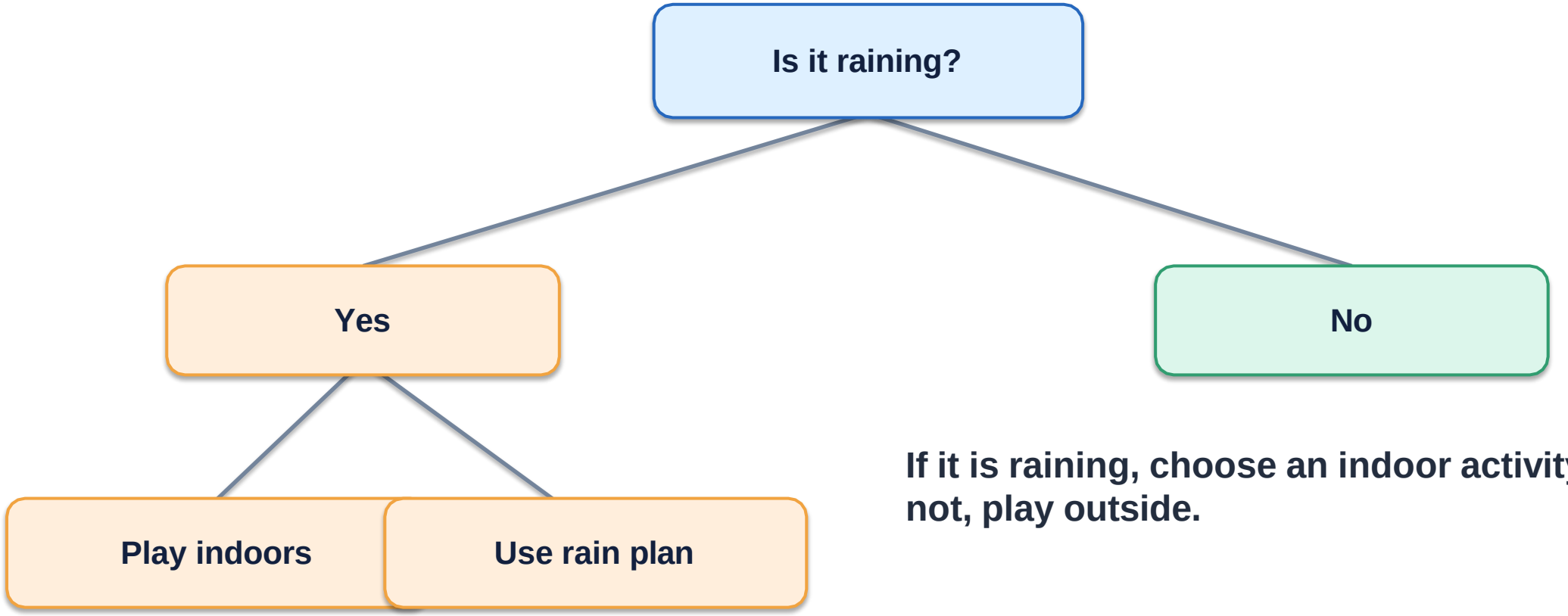
BRANCHES
Possible answers

LEAF NODE
Final prediction

Example uses: spam detection, loan approval, and medical prediction.

2. How Does It Work?

Example: deciding whether to play cricket



If it is raining, choose an indoor activity. If not, play outside.

The tree follows questions and branches until it reaches a decision.

3. Important Concepts

Three ideas used to build a useful decision tree

Entropy

Measures how mixed or uncertain the class labels are.

Lower entropy means the data is more organized.

Information Gain

Measures how much a question reduces uncertainty.

A useful split gives higher information gain.

Pruning

Removes unnecessary branches from a large tree.

It helps reduce overfitting and improves generalization.

Goal: make the tree accurate, simple, and easy to understand.

4. Advantages, Limitations & Conclusion

A quick summary

Advantages

- Easy to understand and explain
- Works with numerical and categorical data
- Useful for classification and decisions

Limitations

- Can overfit the training data
- Small data changes may change the tree
- A very deep tree may be hard to generalize

Conclusion

Decision trees turn data into clear rules. With information gain and pruning, they can support simple, interpretable predictions.

Thank you!